

Technology Stack (Architecture & Tools)

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technology Stack

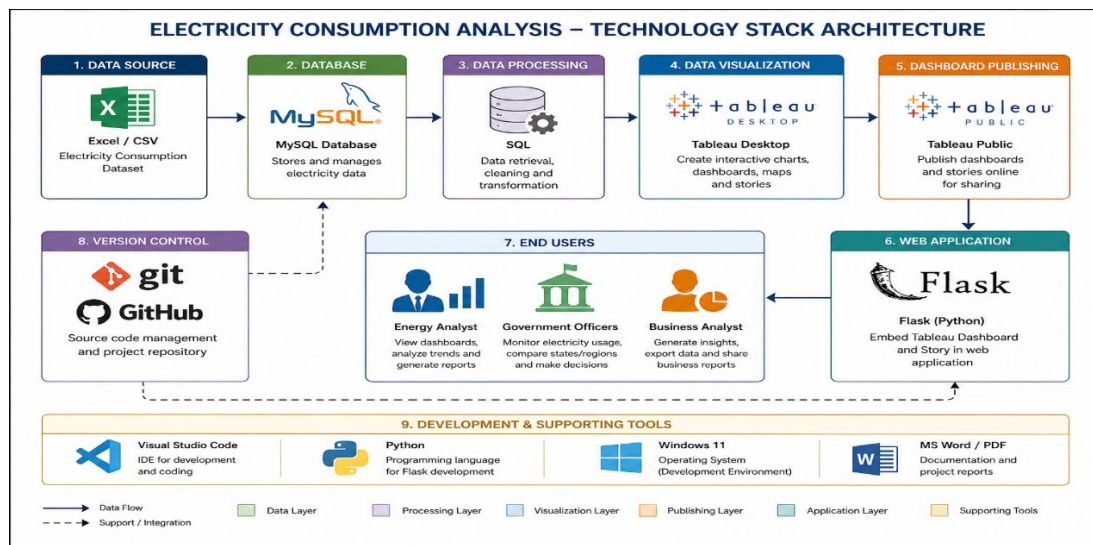
A technology stack describes the collection of software, programming languages, tools, frameworks, and platforms used to develop and deploy the project. The **Electricity Consumption Analysis** project uses MySQL for database management, Tableau Desktop for data visualization, Tableau Public for dashboard publishing, Flask for web application development, and GitHub for version control and source code management.

The selected technologies enable efficient data storage, interactive dashboard creation, web-based visualization, and project documentation.

Technology Stack Table

Layer	Technology Used	Purpose
Dataset	Excel (.xlsx) / CSV	Stores electricity consumption data
Database	MySQL Workbench	Stores and manages electricity data
Query Language	SQL	Data retrieval and analysis
Data Visualization	Tableau Desktop	Create charts, dashboards, maps, and stories
Dashboard Publishing	Tableau Public	Publish interactive dashboards online
Backend Framework	Flask (Python)	Embed Tableau dashboard into web application
Programming Language	Python	Flask web application development
IDE	Visual Studio Code	Application development and coding
Version Control	Git & GitHub	Source code management and project repository
Documentation	Microsoft Word & PDF	Project reports and APSCHE documentation
Operating System	Windows 11	Development environment

Technology Stack Architecture



Advantages of the Selected Technology Stack

- Efficient storage and management of electricity consumption data.
- Interactive dashboards for better visualization.
- Easy comparison of state-wise and region-wise electricity usage.
- Web-based access through the Flask application.
- Secure version control using GitHub.
- Scalable architecture for future enhancements.
- Supports data-driven decision making.
- Easy maintenance and future updates.

Expected Outcome

The selected technology stack provides a complete end-to-end solution for collecting, storing, analyzing, visualizing, and publishing electricity consumption data. It enables users to explore electricity usage patterns through interactive dashboards and supports better planning and decision-making for electricity management.